

Introduction to Blood Sample Collection

Proper blood sample collection is vital. It ensures accurate lab results. This presentation covers phlebotomy steps. Learn about blood collection tubes and their uses. This knowledge is essential for healthcare professionals.



Specimen collection, processing, & preservation



Specimen for Biochemical analysis

- **Specimen** for analysis is the most important element in obtaining reliable test results.

Introduction to Blood Components:

Blood is composed of two main components:

Plasma: This is the liquid part of the blood, making up about 55% of its volume. It's mostly water, but also contains proteins, electrolytes, nutrients, and waste products.

Formed Elements: These are the cells and cell fragments suspended in the plasma. They make up about 45% of blood volume and include:

Red blood cells (Erythrocytes): Carry oxygen.

White blood cells (Leukocytes): Fight infection.

Platelets (Thrombocytes): Help with blood clotting.

Feature	Plasma	Serum
Anticoagulant	Required to prevent clotting. Examples: EDTA, heparin, citrate.	Not required, as blood is allowed to clot.
How to Obtain	Collect blood in a tube with anticoagulant. Centrifuge to separate.	Collect blood in a tube without anticoagulant. Allow to clot. Centrifuge to separate.
Uses	- Coagulation studies	- Immunological tests (antibody detection)
	- Plasma transfusions	- Biochemical analyses (electrolyte, enzyme levels)
	- Measuring certain medications	- Hormone assays
	- Research	- Research

What is Serum and Plasma?

- **Plasma:** Obtained when the blood is collected in a tube with an anticoagulant (e.g., EDTA) and centrifuged. It contains all the proteins, including clotting factors.
- **Serum:** Obtained when blood is left to clot in a tube without anticoagulant either at room temperature or in water bath at 37.0 °C (at least 15 minutes) then blood sample is centrifuged. It lacks clotting factors but contains other proteins, electrolytes, and hormones.

Types of Blood Tests Requiring Serum or Plasma

Explain that certain tests require serum or plasma because they measure specific components

found in these liquids. Here are some examples:

Tests Requiring Serum:

- **Liver Function Tests (LFTs):** Measures enzymes like ALT, AST, and proteins like albumin.
- **Lipid Profile:** Measures cholesterol (HDL, LDL) and triglycerides.
- **Thyroid Function Tests:** Measures TSH, T3, and T4 hormones.
- **Basic Metabolic Panel (BMP):** Measures glucose, electrolytes (sodium, potassium), and kidney function (creatinine, BUN).
- **Serum Protein Electrophoresis:** Analyzes proteins in the blood.

Tests Requiring Plasma:

- 1.Coagulation Tests:** Measures clotting factors (e.g., PT, INR, aPTT).
- 2.Ammonia Level:** Often measured in plasma for liver function assessment.
- 3.Electrolyte Panels:** Sometimes measured in plasma for faster results.
- 4.Renin and Aldosterone Levels:** Hormones measured in plasma for hypertension evaluation and ACTH hormone.

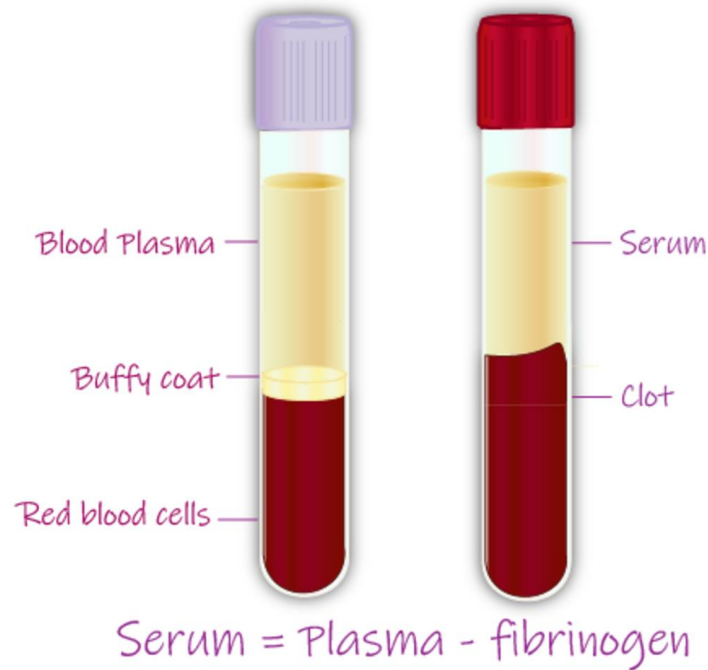
How Serum and Plasma are Collected

- **Serum:** Blood is drawn into a tube **without anticoagulant**, allowed to clot, and then centrifuged to separate the serum.
- **Plasma:** Blood is drawn into a tube **with an anticoagulant, mixed,** and centrifuged to separate the plasma.



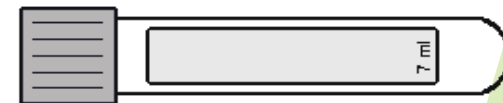
The difference between whole blood, plasma, and serum.

Plasma vs Serum



Common Anticoagulants

- Heparin
- Sodium fluoride
- EDTA
- Oxalates,
- Citrate, Na



•**Heparin:**

- Acts as antithrombin.
- Most widely used, least interference with tests.
- Temporary action, costly, produces blue background in blood smears. Color: Green

•**Sodium Fluoride:**

- Glucose preservative, weak anticoagulant.
- Inhibits glycolytic enzymes, inhibits urease. Color: Gray

•**EDTA (Ethylene Diamine Tetraacetic Acid):**

- Binds calcium.
- Preserves cellular components, unsuitable for Ca & Fe, inhibits ALP & CK. Color: Lavender

•**Oxalates (Na, K, Li, NH₄⁺):**

- Forms insoluble complexes with calcium.
- Inhibits most enzymes, used with fluoride for glucose assays. Color: Gray

•**Iodoacetate:**

- Substitute for NaF as an antiglycolytic agent.
- Does not affect urease, inhibits CK. Color: Gray

•**Sodium Citrate:**

- Chelates calcium.
- Widely used for coagulation studies, unsuitable for Ca, aminotransferase, & ALP assays. Color: Light Blue

How to measure concentration of glucose in blood

A **glucose meter** (or **glucometer**) is a [medical device](#) for determining the approximate concentration of [glucose](#) in the [blood](#).



Follow These 6 Simple Steps To Check Your Blood Glucose Level

1 Wash your hands



2 Load lancing device



3 Insert strip into meter



4 Collect blood sample



5 Apply blood sample



6 Get results in 5 seconds



Normal blood glucose levels

Normal Fasting Blood glucose: 70 - 110 mg/dl [fasting 6-10 hours]

Post prandial Blood glucose : up to 140 mg /dl [2 hours after eating]

Random Blood glucose: up to 180 - 200 mg/dl [after eating 10 min and up]

URINE STRIPS





Step 1

Immerse the reagent area of the strip into the urine sample and take it out quickly.

Step 2

Wait 30 to 60 Seconds



Wait 30 to 60 seconds



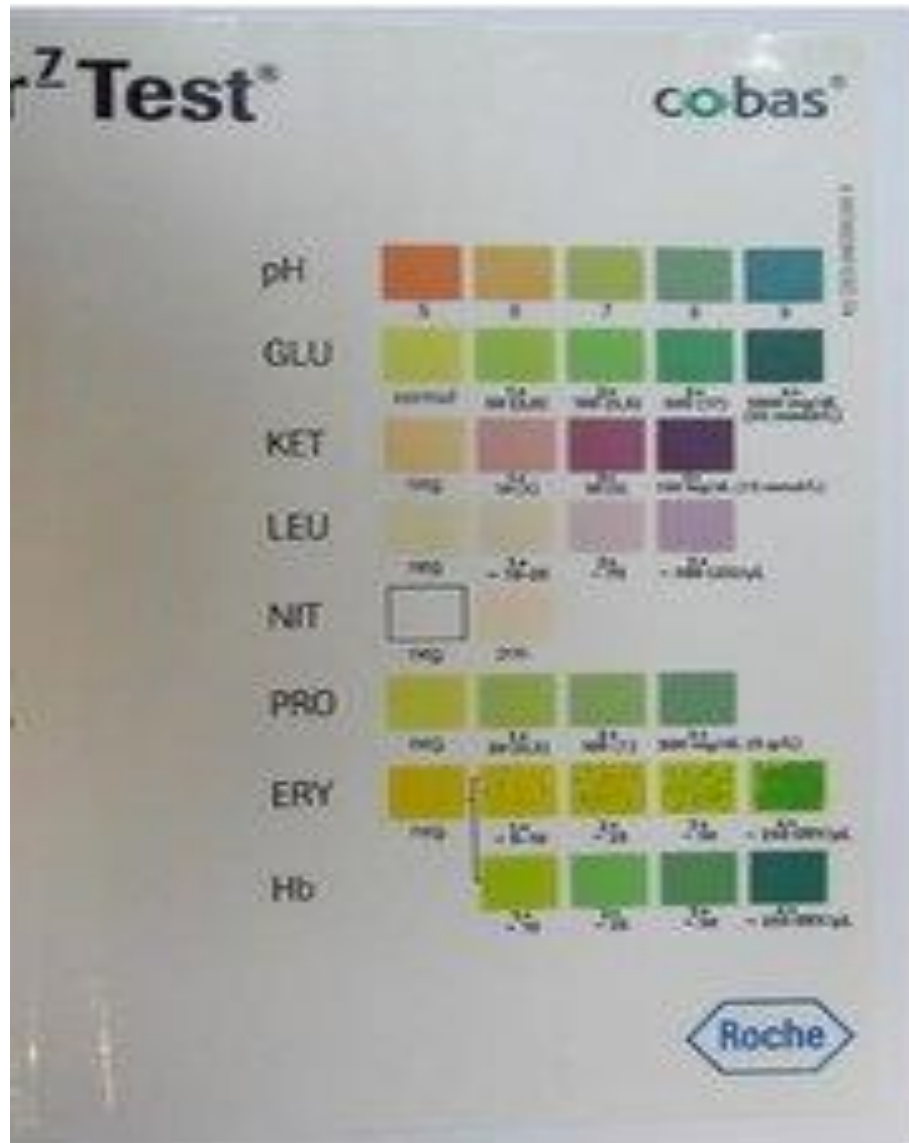
Step 3

Compare the color change of the reagent pads to the corresponding color chart on the bottle label.

Read results according to the chart's time frame for each panel tested.



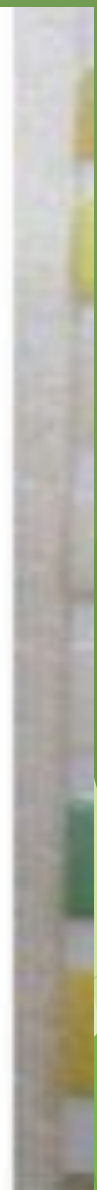




**Combur Test
Strip**



Urine Base



Pro

Leukocytes 120s	Neg.			Trace 15	Small 70	Moderate 125	Large 500	cells/ μ l
Nitrite 60s	Neg.				Positive Any degree of uniform pink color			
Urobilinogen 60s	3.2	Normal	16		32 +	64 ++	128 +++	μ mol/l
Protein 60s	Neg.		Trace ±	0.3 +	1.0 ++	3.0 +++	≥20.0 ++++	g/l
pH 60s	5.0	6.0	6.5	7.0	7.5	8.0	8.5	
Blood 60s	Neg.	Non hemolyzed 10 Trace		Hemolyzed 10 Trace	25 Small	80 Moderate	200 Large	cells/ μ l
Specific Gravity 45s	1.000	1.005	1.010	1.015	1.020	1.025	1.030	
Ketone 40s	Neg.		Trace 0.5	Small 1.5	Moderate 4.0	8.0	Large 16	mmol/l
Bilirubin 30s	Neg.				Small 17	Moderate 50	Large 100	μ mol/l
Glucose 30s	Neg.		5 Trace	15 +	30 ++	60 +++	110 ++++	mmol/l

P.O.C.T

Point Of Care Testing

- Glucose testing
- Urinalysis
- Strep Test
- Pregnancy test
- Fecal Occult Blood
- Cholesterol/Lipid tests
- Coagulation testing
- Rapid Flu testing
- Hemoglobin testing (for anemia)



Glucose



Urine Analysis



Fecal Occult Blood



Rapid Strep



Urine Pregnancy



Coagulation Testing



Cholesterol/Lipids



Hemoglobin



Point Of Care Testing

•**Definition:** POCT refers to medical diagnostic testing performed near the patient away from central laboratory. This allows for quick and convenient results, enabling faster clinical decision-making.

•**Examples:** Common examples include:

- Blood glucose monitoring for diabetes management
- Rapid Strep tests in a doctor's office
- Cardiac marker testing (e.g., troponin) in the emergency department
- Blood gas analysis in intensive care units
- Urine dipstick tests

•Advantages:

- Faster turnaround time for results
- Improved patient convenience and satisfaction
- Potential for reduced length of stay in hospitals
- Enables timely clinical decisions and treatment adjustments

•Disadvantages:

- Potential for errors if not performed correctly
- Quality control and quality assurance can be challenging
- Cost per test may be higher compared to central lab testing
- Connectivity and data management issues

